

HARSH HIRAWAT

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PROFESSIONAL SUMMARY

Full-stack AI engineer and B.Tech CSE student building production LLM, RAG, and backend systems. Experienced with Python, TypeScript, FastAPI/Express, LangChain, LangGraph, Gemini/OpenAI APIs, Pinecone, PostgreSQL, Redis, Docker, AWS, and Cloudflare. Built and deployed real-world AI products including Campus Connect, AskMyNotes, and real-time voice-agent systems, with measurable impact across latency, uptime, and production traffic.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, Go, SQL

AI / Machine Learning: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, MLOps, LLM Fine-Tuning, Model Evaluation, XGBoost

Generative AI & Agents: LangChain, LangGraph, LLM APIs (OpenAI, Gemini, Anthropic, Groq, Mistral), RAG (Pinecone, FAISS, pgvector), Agentic Workflows, Prompt Engineering

Backend Development: FastAPI, Node.js, Express, REST APIs, WebSockets, Microservices, Async Architectures, pytest, Unit Testing

Frontend Development: React, Next.js, Tailwind CSS, Server-Side Rendering (SSR)

Databases: PostgreSQL, Redis, MongoDB

Cloud / DevOps: Docker, AWS, GCP, Cloudflare, CI/CD Pipelines, Git, Containerization

WORK EXPERIENCE

Freelance Full-Stack Developer

Oct 2025 – Present

MindReader Biotech

USA (Remote)

- Architected a subscription-based biotech intelligence platform translating clinical research into actionable investment insights, serving 3+ enterprise clients with secure role-based access and production-grade data workflows.
- Built 5+ high-performance analytics dashboards using Next.js and Tailwind CSS, reducing data-to-insight time by ~35% for end users.
- Deployed containerized microservices using Docker, PostgreSQL, and Redis caching, cutting average deployment time by ~50% via automated CI/CD pipelines.
- Implemented AWS and Cloudflare infrastructure with automated failover, achieving ~99.9% uptime across all production environments.

AI Engineer Intern

Oct 2025 – March 2026

CommKraft

Remote

- Architected a low-latency real-time AI voice agent platform using Gemini API, WebRTC streaming, and asynchronous tool orchestration, supporting 200+ concurrent sessions.
- Built asynchronous LLM orchestration pipeline with LangChain that reduced agent response latency by ~90% (from ~3s to under 300ms), improving real-time conversation quality.
- Designed a state-driven conversation engine with interruption handling, tool execution, and recovery workflows, achieving sub-500ms recovery from failures.
- Implemented human-handoff workflows enabling seamless AI-to-human escalation, reducing escalation handling time by ~60%.

PROJECTS

Evergreen Classic | AI-Powered E-Commerce with WhatsApp Integration

- Link: evergreenclassic.in | [WhatsApp](https://www.whatsapp.com)
- Building a full-stack AI-powered e-commerce platform with WhatsApp Business API integration for conversational commerce, covering order tracking, product discovery, and AI-powered customer support.
- Integrating Gemini API-powered recommendation engine and LangChain agentic pipeline to handle customer queries end-to-end, reducing manual support overhead.
- Architecting backend with FastAPI, PostgreSQL, and Redis Stack; deployed on AWS Lightsail with Docker containerization and automated CI/CD pipelines and proxied with Cloudflare.

Campus Connect | Peer-to-Peer EdTech Mentorship Marketplace

- Links: campus-connect.co.in | [WhatsApp Bot](#)
- Built and deployed a production mentorship marketplace using Next.js, Fastify, PostgreSQL, Redis, Docker, Razorpay, WhatsApp APIs, and Cloudflare.
- Served 636k+ requests, 8.5k+ unique visitors, and 27 GB+ data over a 30-day period with production monitoring and caching.
- Implemented booking, payments, mentor discovery, college comparison, admin operations, async workers, and notification workflows.
- Designed high-availability deployment workflows with PostgreSQL replication, Redis replication, Docker Compose, Cloudflare Tunnel, and automated failover checks.

AskMyNotes | *Corrective RAG Study Companion*

- Links: github.com/GreenHacker420/ASKMYNOTES | askmynotes.greenhacker.in
- Built a citation-backed CRAG system for PDFs and text notes using Next.js, Express, TypeScript, PostgreSQL, Pinecone, Gemini, LangGraph, and Socket.IO.
- Implemented subject-scoped vector retrieval, semantic re-ranking, threshold gating, and grounded answer generation to reduce hallucinations.
- Added real-time streaming chat, voice query support, quiz generation, authentication, and per-user subject isolation.
- Processed 1,000+ document chunks and improved retrieval precision by combining semantic similarity with re-ranking.

ShopControl | *Mobile-First Shop Operations Platform*

- Link: github.com/GreenHacker420/stock-app
- Built a mobile-first operations platform for multi-tenant retail and distribution operations, utilizing a React Native (Expo) frontend and a containerized Node.js (Express) backend.
- Implemented a secure PostgreSQL database layout with Prisma ORM, supporting isolated multi-tenant and multi-shop workflows across sales, inventory, and payment collections.
- Developed asynchronous task workers using BullMQ and Redis for high-throughput background processing, alongside Socket.IO for real-time monitoring and AWS S3 for secure file assets.
- Engineered a custom full-stack Truecaller authentication module (expo-truecaller) for zero-tap mobile verification, with local SQLite caching for offline-first network resilience.

EDUCATION

Newton School of Technology (ADYPU)	Pune, Maharashtra, India
<i>Bachelor of Technology in Computer Science</i>	<i>Aug 2024 – May 2028</i>
• CGPA: 8.86 / 10 • Relevant Coursework: Data Structures and Algorithms, Machine Learning, Distributed Systems, Database Management Systems, Operating Systems, Computer Networks, System Design	

CERTIFICATIONS

AI Engineer Associate Certificate	Feb 2026
<i>School of AI – Udemy</i>	<i>Online</i>
AI Hero: 12-Month Journey from Zero to Expert	Feb 2026
<i>School of AI – Udemy</i>	<i>Online</i>

OPEN SOURCE AND ACTIVITIES

Open Source: Active GitHub contributor with 500+ commits across AI/ML and full-stack repositories at github.com/greenhacker420

Portfolio: Personal developer platform with project showcase and Gemini AI chatbot at greenhacker.in

Key Domains: LLM systems, RAG pipelines, agentic AI, real-time voice agents, WhatsApp automation, cloud-native deployment, full-stack engineering